

GUI Based Voice Identification Using MATLAB



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This voice identification system requires a login password and database of voice samples in your system. Voice samples of the users or authorised persons are first recorded and stored in the computer

memory using a microphone. When the program detects the same voice next time, it displays the object (or user name) being detected and other information. You need a 32-bit operating system with MATLAB software, microphone and webcam.

The voice can be recorded through

normalisation of absolute value of Fourier transform. During this voice-pattern matching process, the program compares with the voice in the database and creates an array of errors or difference outputs. Errors that satisfy the decision rule will be accepted, else rejected. The minimum error object

detected will be considered as recognised information. The block diagram of the voice identification system is shown in Fig. 1.

Software program

The program is developed in MATLAB version R2013a. It has four GUI main panels: password login, speech panel for voice identification, error comparison, and peripheral and property.

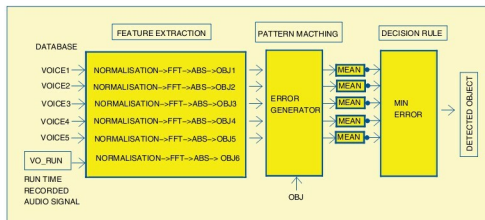


Fig. 1: Block diagram of voice identification system using MATLAB



Fig. 2: Password login panel

the graphical user interface (GUI) program given here. The recorded voice acts as a reference database for the speech identification model. When you run the program and speak in front of the microphone, the program extracts the unique feature from run-time-recorded audio signal using

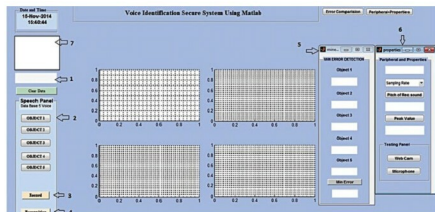


Fig. 3: GUI panel with various sections indicated by arrows

When the demo.m program is run, a screen opens up, prompting the user to enter the password (refer Fig. 2). The password-protected initialisation panel has the ability to report to the administrator when invalid password is attempted five times. The default password is '123456.'

The speech panel is for the selection of voice source from the database. The main parts of the GUI panel are shown in Fig. 3 and the description is given in Table I.

TABLE I
GUI Panel Description

1. Detected object (or user name) after recognition
2. Click to record the voice sample for database – speak single word
3. Run-time voice sample for matching
4. Button for voice recognition
5. Provide error information
6. Peripheral and property info
7. User image display